Escaping Inequality in India: The Role of English-Medium Instruction
(Selected Excerpts)

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What follows are excerpts from my 49-page independent research paper, "Escaping Inequality in India: The Role of English-Medium Instruction." The paper's key points are below.

1. **Question**: Does increasing baseline district-level EMI exposure (percentage of school-going children enrolled in EMI) affect consumption growth over the next decade?
2. **Data sources**: 2007-08/2017-18 microdata and metadata from the National Sample Survey, 35 datasets from the 2001 Census, and 2019-20 GIS shapefiles.
3. **Identification**: District-level 2SLS (pop.-weighted linguistic distance from Hindi IV, state-clustered SEs). Individual-level probit of education selection, design-based SEs.
   - **Key clarification**: State FEs, essential for 2SLS exclusion restriction, trigger extreme multicollinearity. District matching & possible FD-2SLS redesign needed.
4. **Main result**: Strong first-stage ($F = 39.2$), null second-stage estimate (0.2 pp, $p = 0.78$). Estimates are provisional pending district matching that enables state FEs.
5. **Next steps**:
   - Repair 2SLS exclusion restriction with district harmonization changes, FD-2SLS.
   - Redesign models for spatial spillovers, migration, 'bad controls,' heterogeneity.
   - Make response variable robust to local variation in inflation and shocks.

The next ten pages include the following details:

> What is the effect of exposure to English-medium instruction enrollment in 2007-08 on the growth of local consumption from 2007-08 to 2017-18?

First, note the improvements made here over the previous literature. The setting of 2007-2018 may be more relevant today than the settings of similar literature, all of which are closer to the advent of service job offshoring in the 1990s (e.g., Refeque & Azad, 2022). Additionally, note that our response variable here is consumption, not wages. India holds a vast number of informal and self-employed individuals, meaning its labor force varies wildly in the degree to which wages reflect material well-being.[11]

Second, we must highlight that unlike much of this literature, we lack household panel data. We will thus follow previous papers (e.g., Martin et al., 2017) and aggregate data at the district level.[12] More specifically, we will take cross-sectional data from 2007-08 (National Sample Survey Office, 2011) and 2017-18 (National Sample Survey Office, 2022), aggregate each at the district level separately, and match districts across the time periods to construct a district pseudo-panel. District-level variables are developed in Sec. 4.1 and the district matching method is explained in Sec. A.2 .

Our empirical methodology will then proceed in two parts. In Sec. 3, we study the composition of our key "EMI exposure" measure by estimating an individual-level probit of school participation in 2007-08. In Sec. 4, we move to the district level and implement two-stage least squares (2SLS): the weighted average of linguistic distance from Hindi in 2001 will be used to instrument for EMI exposure, onto whose fitted values we regress the growth rate of average consumption.[13] The choice of instrument is discussed in Sec. 4.2.

Our 2SLS results will, at best, reflect the medium-run local general equilibrium effects of EMI exposure. On one hand, by aggregating away all individual or household-level variation, the ecological fallacy renders our 2SLS estimates uninterpretable at these lower levels. Additionally, school-going children in 2007-08 will still be quite young in 2017-18, and many may still be in school (particularly in more developed districts).[14]

With our current methodology, we find that greater exposure to EMI enrollment has a statistically and economically *insignificant* effect on local consumption growth (Sec. 4.4), in line with previous literature on EMI and Indian education quality. Note, however, that our outcome includes households not directly

[11] See Shrivastav & Husain (2026) and Bahl & Sharma (2024) for the frequency and wage effects of informality respectively. Wages may still be more responsive to EMI in the medium-run window we study, however. See Sec. A.1 for more.

[12] A district in India is akin to a county in the United States, but with an average population of 2 million. Table 4 contains more information on district populations.

[13] Future work may incorporate both education selection and EMI exposure as endogenous regressors in a first-difference (FD) 2SLS design. The district-level aggregates of Table 1 may be able to function as education supply-shifting instruments.

[14] Cohort-specific measures of EMI exposure may thus be incorporated into future drafts, as may robustness checks that use school-reported data on medium of instruction to construct EMI exposure—with the aforementioned caveat that schools may lie about having EMI (Lahoti & Mukhopadhyay, 2019; Mathew & Srivastava, 2011). Such surveys may also reveal whether EMI exposure is rigid over time. If so, this could allow us to expand the window of time over which we can interpret our results. Rigidity could also rule out the agglomeration of EMI discussed in Sec. 4.4 as a potential attenuating force behind our results.

3 The Composition of Education Participation

3.1 Model and Variables

We are interested in how consumption growth is affected by EMI exposure (EMIE), which akin to previous literature (Chakraborty & Bakshi, 2016) we define like so:

$$\text{EMIE}_d = 100 \times \frac{\text{Num. children enrolled in EMI}}{\text{Num. children enrolled in school}} \text{ in district } d$$

This measure is effectively a district-level reflection of an intensive margin. To complement our understanding of EMIE in later sections, we would like to understand the extensive margin—enrollment into school—and the variables which affect the probability that any one child i (aged 5-19) appears in our EMIE measure. We thus estimate the following probit model:

$$\begin{aligned}\Pr(\text{Enrolled}_i = 1 \mid X_i,\, Z_{d(i)}) = \Phi\Big(&\beta_0 + \beta_1\,\text{Age}_i + \beta_2\,\text{Female}_i + \beta_3\,\text{HHSize}_i + \beta_4\,\text{Urban}_i \\ &+ \sum_{r\in R}\gamma_r\,\text{Religion}_{ir} + \sum_{s\in S}\delta_s\,\text{SocialGroup}_{is} + \sum_{d\in D}\theta_d\,\text{SchoolDist}_{id} \\ &+ \sum_{f\in F}\psi_f, \text{FatherEduc}_{if} + \sum_{u\in U}\eta_u\,\overline{u}_{d(i)} + \phi\,\overline{\text{EnrollmentCost}}_{d(i)}\Big) \qquad (1)\end{aligned}$$

where

$$\begin{aligned}&R = \{\text{Muslim, Christian, Sikh, Jain, Buddhist, Zoroastrian, None}\},\\ &S = \{\text{Scheduled Tribe, Scheduled Caste, OBC}\},\\ &D = \{\text{1–2km, 2–3km, 3–5km}, \geq \text{5km}\},\\ &F = \{\text{Illiterate; Literate, no school; Literate, school} < \text{primary;}\\ &\qquad \text{Primary, Upper primary, Secondary, Higher secondary, Postsecondary+}\},\\ &U = \{\text{Educ. free available, Tuition waived, Scholarship/Stipend,}\\ &\qquad \text{Textbooks received, Stationery received, Mid-day meal, etc.}\}\end{aligned}$$

and

$$\begin{aligned}\overline{u}_{d(i)} &= \text{Average value of } u \in U \text{ in district } d \text{ amongst enrolled children,}\\ \overline{\text{EnrollmentCost}}_{d(i)} &= \text{Average value of EnrollmentCost in } d \text{ amongst enrolled children.}\end{aligned}$$

There are two kinds of regressors here: those in X_i, the variables unique to child i, and $Z_{d(i)} = \{\overline{u}_{d(i)}$:

$u \in U\} \cup \{\overline{\text{EnrollmentCost}}_{d(i)}\}$, the variables which have been aggregated at the level of the district $d(i)$ where child i lives. Without this aggregation, neither u nor EnrollmentCost, both endogenous to enrollment, would be well-defined for children not in school. Our aggregation can thus be understood as a way of handling missingness; whereas EnrollmentCost_i was missing for 120,062 children (as reflected in Table 1), $\overline{\text{EnrollmentCost}}_{d(i)}$ was missing for 7,109. Observations still missing a relevant variable were then dropped.[17]

The district-level averages can be interpreted as reflections of local supply-side factors, namely the general availability or generosity of educational inputs in a child's local environment. Their marginal effects (see Table 3) may reflect how changes in these district-level characteristics affect the probability a child enrolls in education—or in other words, the elasticity of enrollment with respect to local schooling conditions.[18]

Following Azam et al. (2013), Munshi & Rosenzweig (2006), and Singh (2013), we use father's education to proxy a child's ability before schooling. We use the father despite evidence indicating that the mother's education is more strongly associated with early life human capital development.[19] Taking the average of the mother's and father's education is not possible as education level is recorded by the National Sample Survey Office (2011) as a categorical variable. Future work may attempt to create an index variable for this.

The National Sample Survey Office (2011) do not explicitly identify parent-child relationships. Instead, they only provide each person's relationship to the head of their household. Father's education is thus proxied by the education of the head of the household if they are male (average age here is 49.3); else as the married child of the head if they are male (married men often live with their elderly parents; average age of 40); else as the parent/parent-in-law of the head if they are male (average age of 63.9). We do not include male spouses of the head, whose average age is 11.3 and share of children (aged <18) is 0.813 in our data.[20]

Summary statistics for numeric variables are given in Table 1, and for categorical variables in Table 2.

[17]Three such variables had missing values before dropping: the district-average $\overline{\text{EnrollmentCost}}_{d(i)}$ (7109 missing values), the distance from the nearest primary class (312), and the father's education proxy (67). We find that none of these variables are MCAR (missing completely at random) under logistic regressions with missingness indiciators and adjusted p-values (per Benjamini & Hochberg, 1995). The lowest McFadden pseudo R^2 among the three was a 0.246 on father's education, which had three predictors significant at a level of 0.05 (that is, three variables in the data could predict its missingness with adjusted p-values below the 0.05 threshold). The highest pseudo R^2 was a 0.31 for $\overline{\text{EnrollmentCost}}_{d(i)}$, which had nine significant predictors. Note that prior to this, many other NAs in $\overline{\text{EnrollmentCost}}_{d(i)}$ were handled using codebook-informed aggregations and redefinitions of the variables summed to derive EnrollmentCost_i.

[18]Future work will likely use these district-level averages as instruments for enrollment, an endogenous variable to be included alongside EMIE in a first-difference (FD) 2SLS framework. It may be possible to use this probit as a control function or in a two-stage residual inclusion (2SRI) estimation.

[19]Card (1999) shows that in the US, among people born from before 1920 to 1964, mother's education better predicts school completion among Black people, whereas father's education better predicts for cohorts born 1955-1964. In general, however, he finds mother's education to be the the slightly better predictor of school completion. It's interesting to note that this is a quantitative outcome: one might also expect mother's education to better predict the *quality* of early childhood investment too, with father's education relating more to the quantity of early childhood investment.

[20]This is a clear indication of severe, systematic misclassification error. Further checks on the presence of misclassification imply that the codes for unmarried children and spouses of heads were simply flipped; the average age of male "unmarried children," for example, is 26.9 in the data. Future work will either incorporate the mislabeled "unmarried children" into this proxy chain for father's education, or use the household head's education alone as proxy.

4.2 Instrumental Variable

Our instrument for EMIE_d is LingDistance_d, the average linguistic distance from Hindi of the three most spoken mother tongues in district d in 2001, weighted by the number of speakers then. The mechanism behind this may seem odd: why choose linguistic distance from Hindi as opposed to, say, English? India's large linguistic diversity[27] predisposes it to have large communication costs, large coordination costs, and thus hindered economic development within its borders (Nakagawa & Sugasawa, 2021). To decrease such frictions in the market and in public goods provisioning (Liu & Pizzi, 2016), both Hindi and English have been thrust into the roles of competing domestic lingua franca (Clingingsmith, 2014), meaning Hindi- and English-medium schools can be found across the country (Shastry, 2012, p. 294).

As the 'distance' between one's mother tongue and Hindi increases, we expect the opportunity cost of learning English over Hindi to decrease. Because of this, we also expect individuals with a mother tongue further from Hindi to be more likely to enroll in EMI. "That distance from Hindi induces people to learn English" and even "schools to teach English" is in fact precisely what Shastry (2012, p. 289) claims. If we can then show that local linguistic distance induces exogenous variation in local EMI, we can wield it as an instrumental variable (IV) to isolate EMIE_d's effect on consumption growth. Shastry helps us once more here, evidencing both the relevance (pp. 299-301) and exclusion restriction (pp. 302-305) of LingDistance_d's effect on EMIE_d.

We are currently unable to replicate her justification of the exclusion restriction, however. Her argument centers on a map depicting the geographical balance of her residual variation, made possible despite the distinct regional divide in Figure 3 thanks to state fixed effects. In our case, adding state FEs explodes the condition number of our design matrix to $\kappa = 28730.84$ despite all individual collinearity measures remaining low (every scaled generalized variance inflation factor (GVIF) was below 6.05), with similar results for region FEs to a lesser degree. This almost certainly results from the many-to-many matching used in this paper's district tracking algorithms.[28] Our plan to repair it moving forward is discussed in Sec. A.2.1.

A portion of this near-perfect multicollinearity may simply just be "God's will" (Blanchard, 1987, p. 449): the States Reorganization Act of 1956, for example, aligned state lines with linguistic lines, a precedent which new states have followed since (Sarma, 2025). This possiblity is precisely the motivation for our planned first-difference 2SLS design (Angrist & Pischke, 2008, p. 167). Even if only the data harmonization changes from Sec. A.2.1 are made, future work will still demean each district's linguistic distance by subtracting away its state's linguistic distance.[29]

[27] Perhaps the largest in the world, per one metric from Gurevich et al. (2025).

[28] Many-to-many matching from 2001 to 2007-08 to 2017-18 was used to accurately reflect how real district changes occur as both mergers and partitions. While the intent was accuracy, the effect was degradation: the multiple duplicated rows for 2001 and 2007-08 measures in particular almost surely devastated the rank of the design matrix.

[29] Dimensionality reduction (e.g., principal component analysis, ridge, lasso) and simultaneous equation estimation techniques

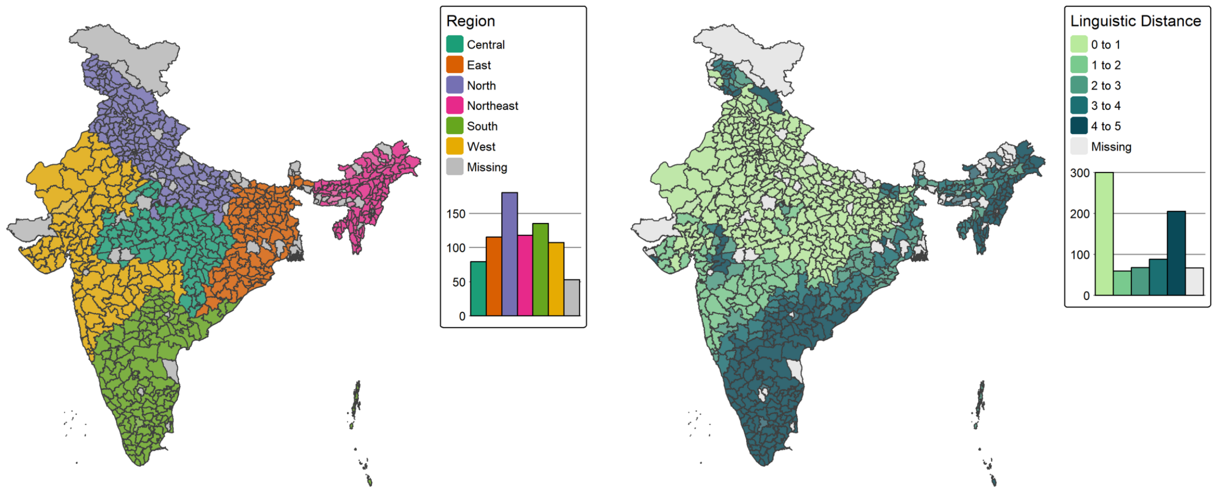

Figure 3: From left to right: regions of India and linguistic distance from Hindi. District-level data, from the 2001 Census of India.

4.3 Results

The results of our first-stage regression, of EMI exposure on linguistic distance, are provided in Table 5. The results of the second-stage regression, where consumption growth is regressed onto the fitted values of EMI exposure, are provided in Table 6. All standard errors in both stages are clustered at the state level.

4.4 Discussion

The interpretations which follow are currently meaningless—the exclusion restriction of our IV is unjustifiable without state fixed effects. For well-conditioned estimation alongside state FEs, future work will demean linguistic distance using the state-level mean and will remove duplicated 2001 and 2007-08 measures by rewriting our district matching algorithms (see Sec. A.2.1). If helpful, a first-difference 2SLS design may also be implemented.

As it stands, however, the F-statistic of our instrument is 39.2, significant at the level of 9.2×10^{-10}, and thus consistent with our identification strategy.[32] The first-stage coefficient on linguistic distance is 2.95, meaning our results are currently consistent with our proposed mechanism (that greater linguistic distance from Hindi induces greater demand for and supply of EMI).

In addition to state fixed effects and additional relevance checks, future drafts will report results from four different specifications: 1) from regressing $\%\Delta\text{Consumption}_d$ on EMIE_d with neither IV nor controls; 2)

[32] This p-value is suspiciously small and likely driven, at least in part, by confounding state/region effects. After adding state FEs, future relevance checks include the partial R^2, statistical tests beyond a Wald with robust covariances, and a jackknife to ensure that leaving out any one state/region does not destroy F.

Table 5: First-Stage Regression: EMI Exposure on Linguistic Distance

	EMI Exposure
	(1)
Linguistic distance	2.945**
	(0.949)
Consumption (2007-08)	0.018***
	(0.005)
Gini of Consumption (2007-08)	−40.452*
	(19.986)
Pct. Urban (ref: Rural)	0.245**
	(0.086)
Average HH size	−0.576
	(2.059)
Dependency ratio × 100	−0.144
	(0.089)
Pct. Female head	−0.415
	(0.555)
Pct. Hindu (ref: Other)	−0.273*
	(0.109)
Pct. Muslim	−0.085
	(0.183)
Pct. Scheduled Tribe (ref: Other)	0.015
	(0.098)
Pct. Scheduled Caste	−0.012
	(0.114)
Pct. OBC	−0.033
	(0.088)
Pct. Small Land-owner (ref: No Land)	0.156*
	(0.078)
Pct. Medium Land-owner	0.167
	(0.118)
Pct. Large Land-owner	0.115
	(0.135)
Pct. Head Educ., Lit.-Primary (ref: Illiterate)	−0.319**
	(0.123)
Pct. Head Educ., Secondary+	0.193
	(0.121)
Intercept	33.076
	(27.314)
Observations	454
R^2	0.630
Adjusted R^2	0.615
Instrument's F-Statistic	39.20***

* $p < 0.05$, ** $p < 0.01$, *** $p < 0.001$
Standard errors clustered by state in parentheses.

from adding the full covariate set, perhaps with the controls and interaction terms discussed in Sec. A.1 plus dimensionality reduction and model selection tests; 3) from regressing $\%\Delta\text{Consumption}_d$ on an $\widehat{\text{EMIE}}_d$

Table 6: Second-Stage Regression: Consumption Growth on EMIE (Fitted)

	Consumption Growth
	(1)
EMIE (fitted)	0.201
	(0.710)
Consumption (2007-08)	−0.104***
	(0.027)
Gini of Consumption (2007-08)	−11.654
	(68.983)
Pct. Urban (ref: Rural)	1.240**
	(0.438)
Average HH size	−5.080
	(6.090)
Dependency ratio × 100	−0.419
	(0.263)
Pct. Female head	−1.807
	(2.167)
Pct. Hindu (ref: Other)	−0.404
	(0.232)
Pct. Muslim	−0.786**
	(0.277)
Pct. Scheduled Tribe (ref: Other)	−0.857***
	(0.238)
Pct. Scheduled Caste	−0.223
	(0.293)
Pct. OBC	−0.615***
	(0.177)
Pct. Small Land-owner (ref: No Land)	−0.169
	(0.313)
Pct. Medium Land-owner	−0.717
	(0.459)
Pct. Large Land-owner	0.567
	(0.395)
Pct. Head Educ., Lit.-Primary (ref: Illiterate)	−0.384
	(0.466)
Percentage. Head Educ., Secondary+	−0.494*
	(0.251)
Intercept	457.536***
	(86.523)
Observations	454
R^2	0.232
Adjusted R^2	0.202

* $p < 0.05$, ** $p < 0.01$, *** $p < 0.001$

Standard errors clustered by state in parentheses.

instrumented by LingDistance_d with no covariates;[33] and finally 4) from estimating our current specification,

[33]Which would allow us to obtain a LATE interpretation without nonparametric estimation or further parametric specification, per Blandhol et al. (2022). Conversely, see Chen (2022) for the conditions under which we can get a conditional LATE.

with both IV and covariates.[34]

In the meantime, as would be expected in our first stage, greater consumption expenditures and urbanization associate with greater EMIE. After controlling for urban/rural settings, it seems that owning at least a small amount of land is associated with greater EMIE too. Also note the -40.45 estimate ($p = 0.043$) on the Gini coefficient: EMIE is lower in more unequal districts. At first, this seems to contradict our literature review in Sec. 1 and its implication that EMI exists within an inequality-amplifying positive assortative matching between households and schools (e.g., Muralidharan, 2019). Larger Gini coefficients, however, often imply longer right tails in income/wealth distributions (Benhabib & Bisin, 2018). We'd thus expect unequal districts to be characterized by a large proportion of poor and a small number of rich households—that is, by a large proportion of students in mother tongue-instruction and a small proportion in EMI. Also reassuring is the small but significant negative result on baseline consumption in the second stage, implying conditional beta-convergence (as described by Vu, 2013) between poorer and richer districts.

The consistency between these results and the literature makes our small and noisy estimate for EMI exposure's effect on consumption growth all the more noteworthy. We find that an ostensibly exogenous percentage point increase in EMIE (i.e., in the percent of a district's school-going children who are enrolled in EMI) associates, at the insignificant p-value of 0.78, with a 0.2 percentage point change in the growth rate of average household consumption expenditures from 2007-08 to 2017-18. Household-level decisions to bet on the human capital and English-language skills possible under EMI do not aggregate into local development when features of household context are accounted for.

This result may be driven by our large urbanization estimate in the second stage (1.24 at $p = 0.005$). Greater EMIE plausibly induces more urbanization: perhaps English skills lead to more service sector firms (Shastry, 2012) and rural-to-urban migration locally (see Sec. A.2.2), or perhaps migration to EMI causes EMI supply to agglomerate and thus even more EMI demanders to migrate. In either case, urbanization would be an outcome of, and thus "bad control" for, EMIE (Angrist & Pischke, 2008, pp. 47–51). Lagged urbanization (see Table 4 for the measure currently used) and the interaction effects of Sec. A.1 should address this.[35]

Migration and spatial spillovers could also explain our small $\widehat{\text{EMIE}}$ coefficient. If EMI helps a person exit to distant urban IT hubs, for example, then the only effect of their EMI on the district they were educated in may be of remittances they send back. It was once widely accepted that migration in India only occurs over short distances, though some new evidence now contests this. Sec. A.2.2 discusses the debate in detail.

[34] Multiple specifications could help us confirm relevance and covariate constructions (e.g., by testing the effect of EMI under different proxies of ability, as Azam et al. (2013) do), but they may be helpful in other ways as well. A positive OLS coefficient on EMIE_d despite a near-zero IV coefficient on EMIE_d, for example, could imply that while EMIE can *predict* local growth, any ostensible causality there is actually driven by unobserved features of households' contexts which correlate with their EMIE.

[35] What they may not address, however, is possible collider bias; see Schneider (2020) for an excellent review.

Interaction terms such as $\text{EMIE}_d \times \text{Urban}_d$ and $\text{EMIE}_d \times \text{TechFirms}_d$ may also be key in identifying the mechanisms by which EMIE can affect a district's growth. Even if households have identical underlying preferences for how to educate their children, the menu of choices available to them (that is, the feasible set over which they optimize) may still be affected by factors far larger than their own education-related decision-making—whether that be by the presence of IT firms nearby, or by the parents' familiarity with and ability to learn more about the Indian education system. To better understand the meaning of the highly significant coefficient on urbanization in Table 6, the urbanization and EMI interaction may be complemented with an urbanization and parental/household head education interaction. The low Muslim/ST/OBC estimates, too, can be studied further by interacting them baseline consumption shares, urbanization, and EMIE itself. And finally, to understand the small coefficients and relative insignificance of the household head's education estimates, we could interact these education levels with urbanization and baseline consumption, or we could even try replacing our baselines with changes, as with a first-difference estimator.

Notice how all actual and proposed regressors in this model reflect baseline exposures. Replacing these with changes and time-varying instruments instead may allow us to use a Bartik-style methodology (e.g., Goldsmith-Pinkham et al., 2020). Linguistic distance, for example, could be turned into a time-varying instrument if interacted by a nation or state-wide shock in EMI supply or demand between 2007-08 and 2017-18. We have yet to identify shocks which have both a plausible effect on EMIE (e.g., changing curricular standards or waves of English training for teachers) and which vary across time alone, not district.

A.2 District Matching and Spatial Autocorrelation

A.2.1 District Matching Method

There are three reasons that district labels vary between 2001 to 2007-08 to 2017-18 in the data: genuine political changes, variation in anglicization, and typos. Fuzzy joins (described below) were used to correct for the latter two.

Political changes, however, were not so easily addressed. The vast majority of such changes were one of five kinds: name changes, clean partitions (one district is partitioned into multiple), clean mergers (multiple districts merge into one), carve-outs (*pieces* of multiple districts combine into one), and boundary shifts (India State Stories, 2024). Without knowing precisely where in its district each household was at the time of sampling, it is not feasible to match districts across carve-outs and boundary shifts. Our district matching method is thus reliant on the assumption that all district changes were either name changes, clean partitions, or clean mergers.

This seems to be precisely the same assumption made by India State Stories (2024) and Jaacks et al. (2020), whose data, despite containing some factual errors and typos, was the basis of our district matching method. To justify this assumption we turn to H. Kumar & Somanathan (2016), who up until 2001 are able to track the *proportion* of each district's population that was allocated into a new district as a result of district changes. They label these changes as one of two types: clean partitions and non-partitions (which would include name changes, clean mergers, carve-outs, and border shifts). By extracting their results using Tabula (Aristarán et al., 2018), we know that 93.4% of non-partitions were either name changes or transfers of uninhabited land (H. Kumar & Somanathan, 2016). These district changes are plotted in Figure 4.

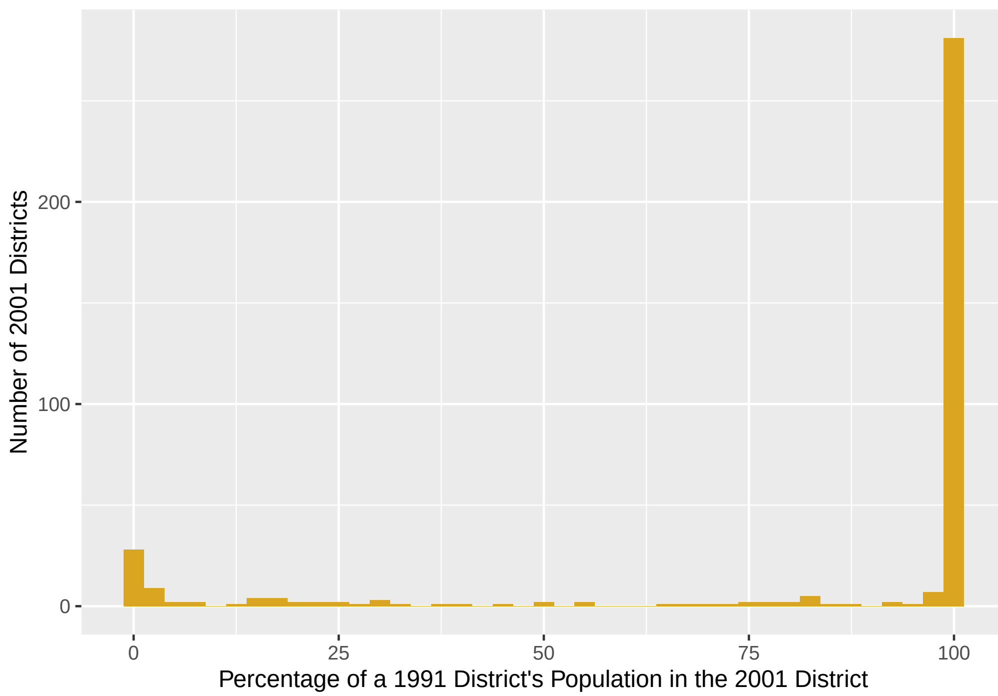

Figure 4: Number of 2001 districts which absorbed a percentage of a 1991 district's population via name change, clean merger, carve-out, or border shift. Data from Kumar & Somanathan (2016).

We thus assume no district changes were carve-outs or border shifts. To match districts, we ran a hierarchical, cascading sequence of iterated fuzzy joins. Data from 2001, 2007-08, and 2017-18 is first linked to the closest year in the district trackers of India State Stories (2024) and Jaacks et al. (2020). Districts are then matched if they meet a certain string/phonetic distance threshold. Matches were made in the following order:

1. `soundex` $= 0$, for phonetically equivalent variations in anglicization (e.g., "Baleswar" and "Balasore").

2. `qgram` distance $= 0$, for rearrangements of bigrams (e.g., "East Godavari" and "Godavari East").

3. Jaro-Winkler distance ≤ 0.15, for respellings and vowel swaps with 0.85 degrees of similarity.

4. Full Damerau-Levenshtein distance ≤ 2, for ≤ 2 insertions, deletions, substitutions, and transpositions.